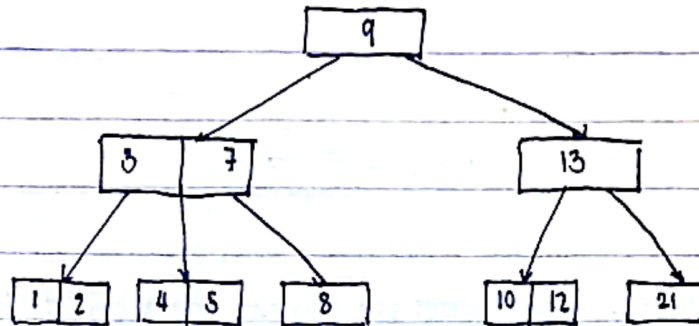


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- B-TREE OF ORDER 4 (INSERT: 5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8)

FINAL TREE

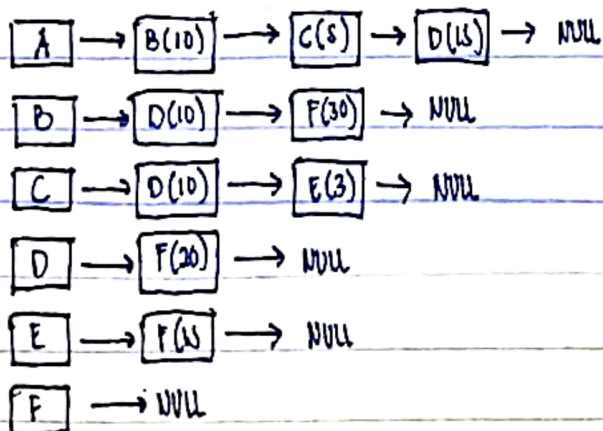


- CONSTRUCT THE FOLLOWING GRAPH REPRESENTATION FROM GIVEN DIRECTED GRAM ABOVE

→ ADJACENCY MATRIX

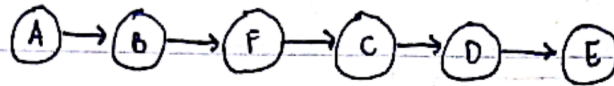
	A	B	C	D	E	F
A	0	10	5	15	0	0
B	0	0	0	10	0	30
C	0	0	0	10	3	0
D	0	0	0	0	0	20
E	0	0	0	0	0	15
F	0	0	0	0	0	0

→ ADJACENCY LIST

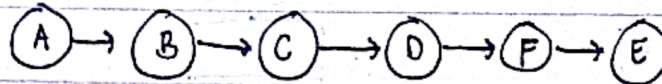


- SHOW THE DEPTH-FIRST & DEPTH FIRST TRAVERSAL.

→ DFS



→ BFS



- FIND ALL VALID PATH SEQUENCE. FIND THE SHORTEST PATH SEQUENCE THROUGH SUM OF THE WEIGHTS OF THE EDGES INVOLVED IN THE PATH

PATH	EDGES	T. WEIGHT
A → B → F	10 + 30	40
A → D → B → F	15 + 10 + 30	55
A → D → F	15 + 20	35
A → C → D → B → F	5 + 10 + 10 + 30	55
A → C → D → F	5 + 10 + 20	35
A → C → E → F	5 + 3 + 15	23

SHORTEST PATH

A(5) → C(3) → E(15) → F ⇒ 23